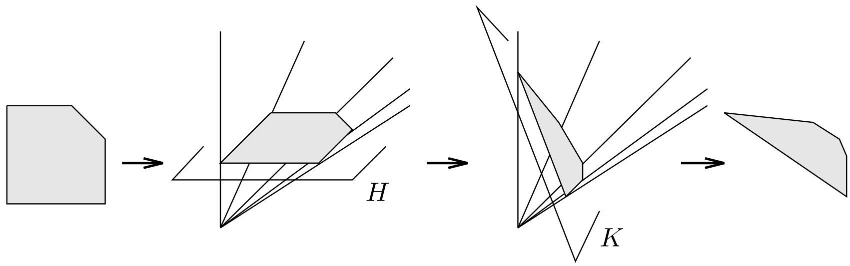
polytopes are examples of simplicial polytopes. Note that in their "usual" symmetric version as regular polytopes, the dodecahedron and the icosahedron do *not* have rational coordinates. (In fact, as regular polytopes they cannot be represented in rational coordinates!)

Any d-polytope that is both simple and simplicial is a simplex if $d \geq 3$ — this was covered in Exercise 0.0. To see this, let P be simple, and consider a vertex $\boldsymbol{v}$ of P. This vertex is in exactly d facets, which are all simplices. Looking at the vertex figure, we see that $\boldsymbol{v}$ is on d edges, and the d vertices $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_d$ adjacent to $\boldsymbol{v}$ are also adjacent to each other: here we use the condition $d \geq 3$. This means, since the same argument could start at $\boldsymbol{v}_i$, that there are no other vertices than $\boldsymbol{v}, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_{i-1}, \boldsymbol{v}_{i+1}, \ldots, \boldsymbol{v}_d$ adjacent to $\boldsymbol{v}_i$. Thus the vertex set of P is $\{\boldsymbol{v}, \boldsymbol{v}_1, \ldots, \boldsymbol{v}_d\}$, and P is the simplex spanned by this set.

2.6 Appendix: Projective Transformations

Although linear transformations are our main tool to "put polytopes where we need them," it is sometimes convenient to use more general transformations, which allow us even to "adjust the shape of a given polytope," known as *projective transformations.*

We can describe projective transformations in a very simple way with the tools of this lecture (in particular, without construction of projective space and use of projective geometry). For this, we proceed as follows. Given a polytope $P \subseteq \mathbb{R}^d$, we embed it into an affine hyperplane $H \subseteq \mathbb{R}^{d+1}$, and construct homog$(P)$, the homogenization of P. By construction, this is a pointed cone. Now we cut this cone by a different hyperplane $K \subseteq \mathbb{R}^{d+1}$, which is then identified with $\mathbb{R}^d$ by an affine map.



Here K is required to be an *admissible hyperplane*: an affine hyperplane that intersects every ray in homog(P) that starts at $\mathbf{0}$. Under this condition, we get a new polytope $P' \subseteq \mathbb{R}^d$, which is affinely isomorphic to $K \cap \text{homog}(P)$, and thus combinatorially equivalent to P:

$$L(P) \ \cong \ L(\text{homog}(P)) \ \cong \ L(P').$$

Briefly, the geometric procedure can be described as homogenization (embedding into an affine hyperplane) followed by dehomogenization (with respect to a new affine hyperplane).

We now derive formulas that directly describe the map $f : P \longrightarrow P'$ in d-dimensional space. For this, let

$$H \;=\; \{\begin{pmatrix}\boldsymbol{x}\\1\end{pmatrix} : \boldsymbol{x} \in \mathbb{R}^d\} \;=\; \{\begin{pmatrix}\boldsymbol{x}\\x_{d+1}\end{pmatrix} \in \mathbb{R}^{d+1} : x_{d+1} = 1\} \;\subseteq\; \mathbb{R}^{d+1}$$

and

$$K \;=\; \{\begin{pmatrix}\boldsymbol{x}\\x_{d+1}\end{pmatrix} \in \mathbb{R}^{d+1} : \boldsymbol{a}\boldsymbol{x} + a_{d+1}x_{d+1} = 1\}.$$

The hyperplane K is admissible if and only if $(\boldsymbol{a}, a_{d+1})\binom{\boldsymbol{v}}{1} > 0$ for all vertices $\boldsymbol{v} \in \operatorname{vert}(P)$. We map K back to $\mathbb{R}^d$ via an affine map

$$\pi : K \longrightarrow \mathbb{R}^d, \qquad \begin{pmatrix}\boldsymbol{x}\\x_{d+1}\end{pmatrix} \longmapsto B\boldsymbol{x} + x_{d+1}\boldsymbol{z} + \boldsymbol{z}'.$$

This map is an isomorphism $\pi : K \cong \mathbb{R}^d$ if and only if

$$\det\begin{pmatrix}B & \boldsymbol{z}\\ \boldsymbol{a} & a_{d+1}\end{pmatrix} \neq 0.$$

This means that for $a_{d+1} \neq 0$ we could take $B = I_d$, $\boldsymbol{z} = \boldsymbol{z}' = \boldsymbol{0}$, and thus $\pi(\begin{pmatrix}\boldsymbol{x}\\x_{d+1}\end{pmatrix}) = \boldsymbol{x}$.

Putting these elements together, we get the projective transformation in formulas as

$$\boldsymbol{x} \in P \;\mapsto\; \begin{pmatrix}\boldsymbol{x}\\1\end{pmatrix} \in H \;\mapsto\; \frac{1}{\boldsymbol{a}\boldsymbol{x} + a_{d+1}}\begin{pmatrix}\boldsymbol{x}\\1\end{pmatrix} \in K \;\mapsto\; \frac{B\boldsymbol{x} + \boldsymbol{z}}{\boldsymbol{a}\boldsymbol{x} + a_{d+1}} + \boldsymbol{z}' \in \mathbb{R}^d.$$

Thus a projective transformation acts on P as a rational linear map, and we get for P' the general formula

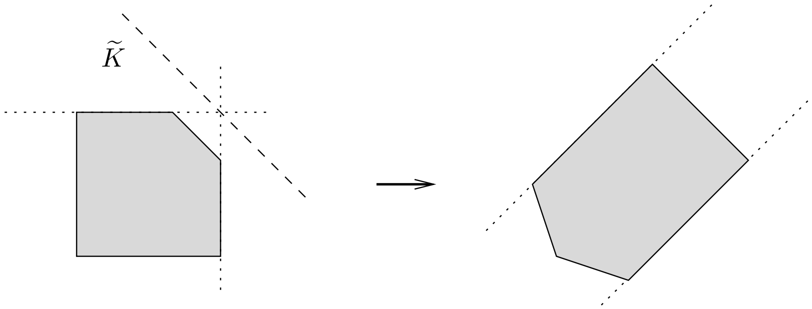
$$P' \;=\; \left\{\frac{B\boldsymbol{x} + \boldsymbol{z}}{\boldsymbol{a}\boldsymbol{x} + a_{d+1}} + \boldsymbol{z}' : \boldsymbol{x} \in P\right\},$$

under the conditions that

$$\det\begin{pmatrix}B & \boldsymbol{z}\\ \boldsymbol{a} & a_{d+1}\end{pmatrix} \neq 0,$$

and that $\boldsymbol{a}\boldsymbol{v} + a_{d+1} > 0$ for all $\boldsymbol{v} \in \operatorname{vert}(P)$.

Geometrically, the transformation has the following effect. The map is defined on the interior of the halfspace $\widetilde{K}^+ = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{a}\boldsymbol{x} + a_{d+1} \geq 0\}$. It maps subspaces that meet the interior of $\widetilde{K}^+$ to subspaces of the same dimension. The hyperplane $\overline{K}$, which bounds the halfspace, is "moved to infinity" by the projective transformation. Hence, lines that intersect in the interior of $\widetilde{K}^+$ are mapped to intersecting lines, while lines that meet on the boundary hyperplane $\widetilde{K} = \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{a}\boldsymbol{x} + a_{d+1} = 0\}$ of $\widetilde{K}^+$ are mapped to parallel lines.



It is certainly instructive to get a good book on projective geometry — you might try Garner [224] for the basics, or classical treatments like Veblen & Young [552] and Hodge & Pedoe [276] for more — and study its definition and description of projective transformations, and try to match it with the one given here.

Understanding projective transformations is absolutely necessary if you work more with polytopes, even if we will not use them much in the following lectures. Nevertheless, the reader will recognize them at times (used explicitly and implicitly).

Projective transformations are used for "preprocessing," to get a polytope into the shape to apply certain procedures, without changing the combinatorial structure. For this, it is (almost) never necessary to go back to the formulas: it suffices to see geometrically that a projective transformation with certain properties exists. Two such applications are described in Exercises 2.17 and 2.18 — both will be useful later.

Notes

All the basic facts about polarity, the face lattice, and the various parts of the representation theorem and their proofs are classical, due to Farkas, Weyl, Minkowski, Carathéodory, Motzkin, Kuhn, and others. Again we refer to Grünbaum [252] and Schrijver [484] for the history. Anyway, "history will teach us nothing" (Sting).

(This was a message from our No Comment department.)